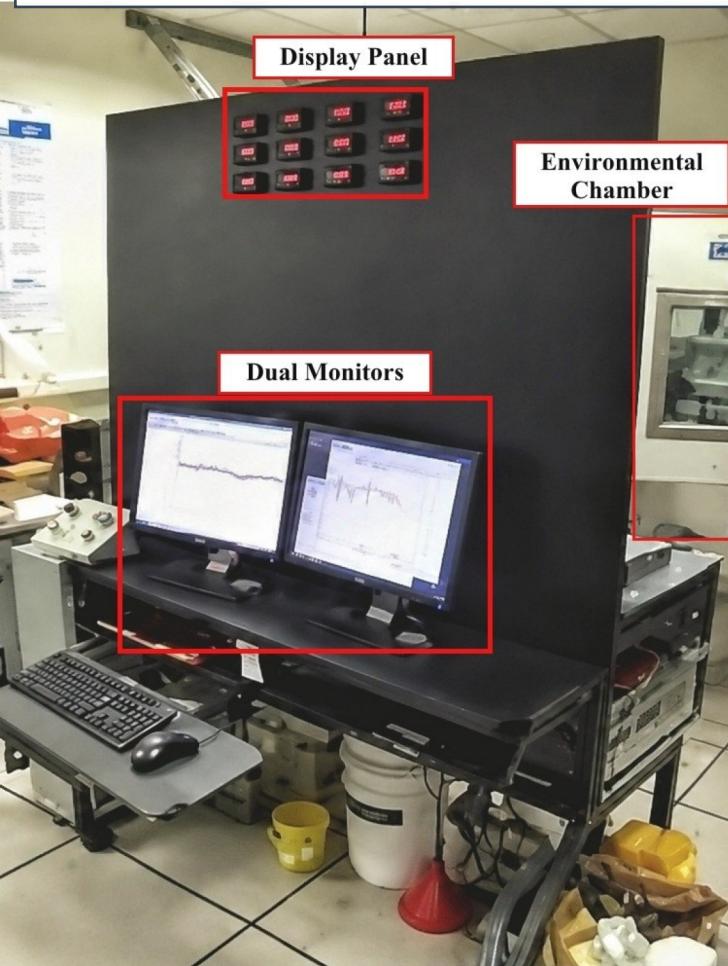
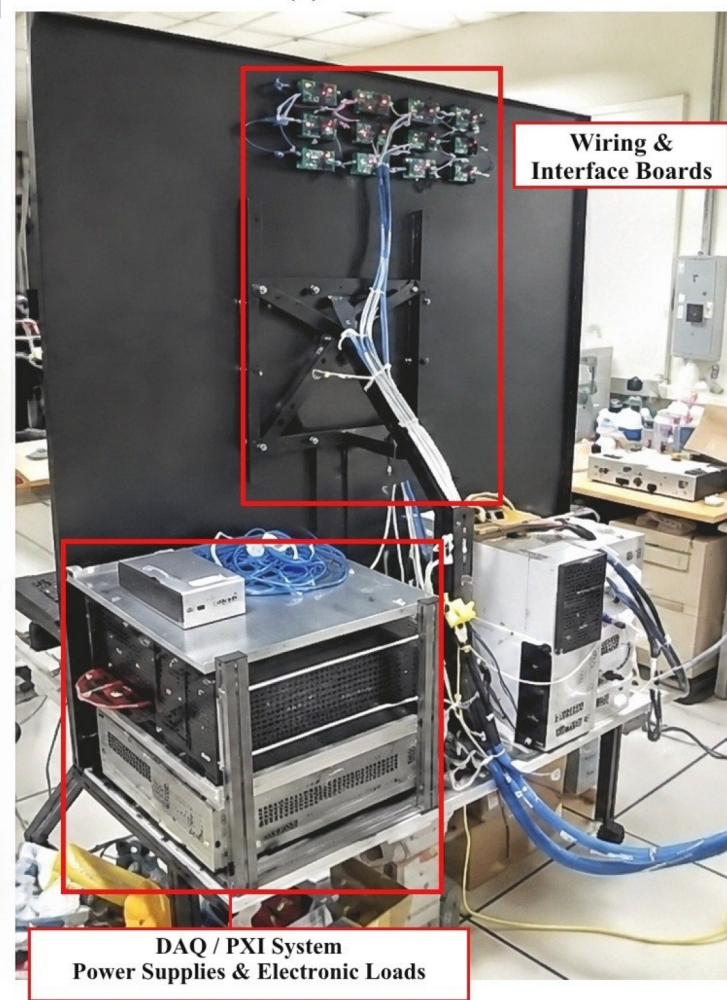


(a) NASA Battery Aging Experimental Bench



(b) Rear View



(b) Data Acquisition Procedure (NASA Ames PCoE)

Step 1: Thermal Chamber

- Commercial 18650 Li-ion cells (2.0 Ah)
- Tenney environmental chamber
- Fixed ambient: 4 °C, 24 °C, or 43 °C

Step 2: CC-CV Charging

- Programmable DC power supply
- CC mode: 1.5 A charging to 4.2 V
- CV mode: 4.2 V until current ≤ 20 mA

Step 3: Load Discharging

- 4-channel programmable DC load
- Constant-current discharge (2 A / 4 A)
- Cycled to cutoff voltage (2.0-2.7 V)

Cycle Repeat & Degradation

Step 5: PXI DAQ & MATLAB Raw Storage

- National Instruments PXI chassis continuous acquisition (~ 10 Hz)
- Automated experiment control & raw cycle generation (.mat files)
- Repeated until 30% End-of-Life (EOL) capacity fade (2.0 Ah $\rightarrow$ 1.4 Ah)

Step 4: Multimodal In-Situ Sensing & EIS

- Synchronous sensing: Terminal voltage (V) & current (I)
- Type-K thermocouple mounted on cell body for temperature (T)
- Periodic Electrochemical Impedance Spectroscopy (0.1 Hz– 5 kHz)

(c) Evaluated NASA Li-ion Cells & Experimental Operating Parameters

Cell ID	Ambient Temp.	Discharge Current	Cutoff Voltage	Cycles (n)	SOH Range (Fade)	Experimental Role
B0005	24 °C (Room)	2.0 A (1.0 C)	2.7 V	168	1.000 $\rightarrow$ 0.714	Training (Seen Cell)
B0006	24 °C (Room)	2.0 A (1.0 C)	2.5 V	167	1.000 $\rightarrow$ 0.583	Training (Seen Cell)
B0007	24 °C (Room)	2.0 A (1.0 C)	2.2 V	168	1.000 $\rightarrow$ 0.758	Training (Seen Cell)
B0018	24 °C (Room)	2.0 A (1.0 C)	2.5 V	132	1.000 $\rightarrow$ 0.723	Test (Unseen Cross-Cell)
B0029	43 °C (High)	4.0 A (2.0 C)	2.0 V	40	1.000 $\rightarrow$ 0.950	Training (Seen Cell)
B0030	43 °C (High)	4.0 A (2.0 C)	2.2 V	40	1.000 $\rightarrow$ 0.944	Training (Seen Cell)
B0031	43 °C (High)	4.0 A (2.0 C)	2.5 V	40	1.000 $\rightarrow$ 1.000	Training (Seen Cell)
B0032	43 °C (High)	4.0 A (2.0 C)	2.7 V	39	1.000 $\rightarrow$ 0.960	Test (Unseen Cross-Cell)
B0053	4 °C (Cold)	2.0 A (1.0 C)	2.2 V	53	1.000 $\rightarrow$ 0.945	Train: cyc 0-36 (n=37) / Test

Summary: Commercial 18650 Li-ion cells (LiCoO₂/Graphite, C_{nom} = 2.0 Ah, V_{nom} = 3.7 V). Total cycles: 847 (Train: 660, Test: 187). Resampled